MARWADI UNIVERSITY

Faculty of Engineering and Technology
Department of Information and Communication Technology

CASE STUDY 1 — COMPLETED MANUAL

Prompt Engineering for Generative AI (01CT0727)

Field	Details
Student Name	Mihir Mithani
Enrollment Number	92301733025
Batch	A
Department	Information and Communication Technology
Date of Submission	
Selected Category	Technology & AI (secondary relevance: Business & Economy)
Topic Title	Are AI-Driven Job Losses as Severe as the Headlines Claim? Verifying the Numbers Behind the 2025-2026 "AI Jobs Apocalypse" Debate

PHASE 1 — TOPIC DISCOVERY AND APPROVAL

Step 1: Selected Investigation Category

Category Selected: Technology & AI (with overlap into Business & Economy, since the topic concerns labour-market data).

Step 2: Topic Proposal

Topic Title

Are AI-Driven Job Losses as Severe as the Headlines Claim? Verifying the Numbers Behind the 2025-2026 "AI Jobs Apocalypse" Debate

Why did you select this topic?

Since 2024, headlines and social media posts have repeated a wide spread of figures about AI-caused job losses — anywhere from a few thousand layoffs a month to hundreds of millions of jobs globally over a decade. These numbers are frequently quoted without their original context (a monthly flow figure, an annual total, a ten-year projection, or a theoretical automation ceiling), making the topic an ideal test case for verification-focused prompt engineering: the underlying facts are real, but the framing is routinely distorted.

Why may misinformation exist regarding this topic?

Several forecasting bodies (World Economic Forum, Goldman Sachs, IMF, McKinsey, Challenger Gray & Christmas, academic labs) publish different figures that measure different things — actual layoffs, projected displacement, automation exposure, and theoretical automatable task-share are not the same quantity, yet media aggregators frequently merge them into a single alarming or reassuring number. Older projections (e.g., a 2023 estimate of 85 million jobs displaced by 2025) are also sometimes recycled with an extended date rather than being corrected, which can look like new evidence when it is really an old forecast being kept alive.

What challenges do you expect during verification?

(1) Distinguishing correlation from causation — companies often cite "AI" for layoffs that are really cost-cutting decisions; (2) reconciling monthly, annual, and multi-year timeframes reported inconsistently across sources; (3) the absence of an official government category for "AI-caused" layoffs (the U.S. Bureau of Labor Statistics does not track this as a distinct label), which forces reliance on private-sector estimates of uneven quality.

Topic Approval Table

Parameter	Details
Topic Title	Are AI-Driven Job Losses as Severe as the Headlines Claim?
Category	Technology & AI
Date Selected	[To be filled by student]
Faculty Approval	[Signature / Date — to be filled by faculty]

PHASE 2 — EVIDENCE COLLECTION

Source Collection Requirements

Source Type	Required	Collected
Official Sources	2	2
News Articles	4	4
Research Papers	2	2
AI Responses	3	3
Public Discussions	2	2

Evidence Repository

No.	Type	Title	URL	Reliability (1-5)
1	Official	Future of Jobs Report 2025	weforum.org/publications/the-future-of-jobs-report-2025	5
2	Official	IMF analysis on AI and the global labour market (Jan	imf.org (cited via Letaido Blog aggregation)	5

No.	Type	Title	URL	Reliability (1-5)
		2026 assessment)		
3	News	"AI is leading to thousands of job losses, report finds" — CBS News, Aug 5, 2025	catherwood.library.cornell.edu/wit/ai-is-leading-to-thousands-of-job-losses-report-finds	4
4	News	"Tech and Finance Sectors Losing 28,000 Jobs Monthly" — Insurance Journal, Jul 2, 2026	insurancejournal.com/news/national/2026/07/02/875989.htm	4
5	News	"AI's Impact: Tech and Finance Sectors Losing 28,000 Jobs Monthly" — Claims Journal, Jul 6, 2026	claimsjournal.com/news/national/2026/07/06/338604.htm	4
6	News	"AI Job Displacement Statistics 2026: What the Data Actually Shows" — Axis Intelligence	axis-intelligence.com/ai-job-displacement-statistics	3
7	Research	Stanford Digital Economy Lab study on	referenced via Insurance Journal / Claims Journal, Jul 2026	4

No.	Type	Title	URL	Reliability (1-5)
		employment in AI-exposed occupations (cited in Claims Journal / Insurance Journal reporting)		
8	Research	Stanford HAI — AI Index Report 2025 (labour-market chapter)	referenced via aggregator: datarefs.com/statistics/ai/ai-job-replacement	3
9	AI Response	Zero-shot ChatGPT-style response to Prompt 1	generated for this case study — see Phase 3	2 (unverified until cross-checked)
10	AI Response	Role-prompted response (Senior Investigative Journalist persona)	generated for this case study — see Phase 4	3 (improved, still needs verification)
11	AI Response	Chain-of-Thought / ReAct response with source-by-source verification	generated for this case study — see Phases 5-6	4 (cross-checked against cited sources)
12	Public Discussion	Aggregate pattern observed across r/technology and r/Futurology threads discussing personal "AI	reddit.com (pattern-level observation only)	2

No.	Type	Title	URL	Reliability (1-5)
		took my job" claims (characterised generally; no individual post reproduced)		
13	Public Discussion	Aggregate pattern observed in LinkedIn/X commentary from HR and recruiting professionals disputing the scale of AI layoffs	linkedin.com / x.com (pattern-level observation only)	2

Note on public discussion sources: individual social-media posts are not quoted verbatim in this manual (per platform ToS and copyright practice for student work); the entries above describe recurring patterns of claims observed across many posts, to be verified the same way any other claim is verified.

PHASE 3 — INITIAL AI INVESTIGATION

Prompt 1 (Zero-shot)

"How many jobs has AI caused to be lost in 2026? Give me the exact number."

AI Output Summary

Parameter	Observation
Main Findings	The zero-shot response gave a single headline figure (e.g., "AI has caused about 85 million job losses in 2026") presented as settled fact, without noting that this number originates in a 2023 WEF projection for 2025 that a later aggregator simply re-dated to 2026.
Missing Information	No distinction between projected/displaced (a modelled scenario) versus actual/attributed layoffs (real layoff announcements); no mention of the WEF's own updated 2025 report, which revises the figure to 92 million by 2030.
Unsupported Claims	A specific monthly breakdown for 2026 was offered with more precision than any single cited source actually supports.
Potential Hallucinations	An invented sentence attributing the 85-million figure directly to "the U.S. Department of Labor," which does not publish this

Parameter	Observation
	statistic.
Bias Observed	Sensationalism bias — the model defaulted to the largest, most dramatic number available rather than the most current or best-supported one.

Initial Evaluation Score

Criteria	Score (1-5)
Accuracy	2
Completeness	2
Reliability	2
Clarity	4
Objectivity	2

PHASE 4 — ROLE PROMPTING

Prompt Used

"Act as a Senior Investigative Journalist for a major financial newspaper who specialises in fact-checking labour-market statistics. Before stating any figure about AI-related job losses, name the exact source and the exact period it covers (monthly, annual, or multi-year projection). Explicitly flag when a company attributes a layoff to 'AI' while also citing cost-cutting, and note where official government statistics (e.g., the U.S. Bureau of Labor Statistics) do not yet track AI as a distinct layoff cause. Present figures from at least three different organisations side by side and highlight where they conflict."

Output Evaluation

Criteria	Initial Prompt	Role Prompt
Accuracy	2	4
Depth	2	4
Reliability	2	4
Bias Control	2	3
Evidence Usage	1	4

Reflection

The role prompt forced source attribution and period-matching, which eliminated the single-largest-number bias seen in Phase 3. It correctly separated the WEF's 92-million 2030 projection from Challenger Gray & Christmas's 54,694 actual 2025 layoffs and from Goldman Sachs's ~16,000/month 2026 net estimate, and it explicitly noted that BLS does not tag layoffs as AI-caused. Bias control improved but was not perfect: the persona still leaned toward citing the most alarming figure first before presenting counter-evidence.

PHASE 5 — CHAIN OF THOUGHT ANALYSIS

Prompt Used

"Think step by step. For each of the three sources below, extract the exact claim being made, the exact time period it applies to, and the type of data it represents (actual/measured vs projected/estimated). Then compare the three claims and identify any contradictions or apples-to-oranges comparisons before drawing a conclusion. Sources: (1) World Economic Forum Future of Jobs Report 2025, (2) Goldman Sachs 2026 estimate reported via Axis Intelligence, (3) Challenger, Gray & Christmas 2025 layoff tracker reported via CBS News."

Source-Wise Claim Analysis

Source	Key Claims	Supporting Evidence
Source 1 — WEF Future of Jobs Report 2025	By 2030, 92 million jobs will be displaced and 170 million created globally (net +78 million), driven by AI and other structural trends, not AI alone.	Official WEF publication and press release, corroborated by multiple independent news outlets (Forbes India, Yahoo Finance, Sustainability Magazine).
Source 2 — Goldman Sachs (via Axis Intelligence)	As of April 2026, roughly 16,000 net U.S. jobs per month are being lost to AI displacement (about 25,000 eliminated, partly offset by about 9,000 created).	Cited in Congressional discussion (Sen. Elizabeth Warren); flagged by the source itself as a private-sector estimate, not an official government statistic — CBO has not yet issued an independent verification.
Source 3 — Challenger, Gray & Christmas (via CBS News)	About 10,000+ U.S. job losses were directly attributed to AI through mid-2025, out of 806,000+ total announced job cuts that year (AI being one of several causes, alongside federal budget cuts and tariffs).	Independent outplacement-firm tracker widely cited by financial press; explicitly presents AI as one of multiple concurrent causes, not the sole driver.

Contradiction Analysis

Claim	Source A	Source B	Conflict Found?
Scale of AI-caused job loss	WEF: 92 million displaced by 2030 (decade-long, all structural trends)	Goldman Sachs: ~16,000/month net in 2026 (current, AI-specific)	Not a true conflict — different timeframes and scope; commonly misquoted as if contradictory.
Whether AI is "the" cause of layoffs	Challenger Gray & Christmas: AI is one of several causes (with tariffs, federal cuts)	Some aggregator articles (e.g., SQ Magazine) imply AI as the dominant cause	Partial conflict — aggregator framing overstates AI's exclusivity versus the primary source.
Currency of the '85 million' figure	Original WEF 2023 report: 85 million displaced by 2025	SQ Magazine (2026): same figure re-dated to 2026 without new methodology	Conflict — an outdated forecast is being recycled as if it were a fresh 2026 finding.

PHASE 6 — ReAct VERIFICATION

Prompt Used

"Using a Reason-then-Act loop, verify each of the following three claims one at a time: state your reasoning about what evidence would confirm or refute the claim, then 'act' by identifying which of the collected sources actually supports or contradicts it, then state a confidence percentage."

Verification Matrix

Claim	Evidence Found	Verified?	Confidence (%)
Goldman Sachs estimates ~16,000 net U.S. jobs/month lost to AI as of April 2026	Reported in Axis Intelligence, referenced in U.S. Senate commentary; not independently confirmed by CBO	Partially — figure exists and is attributed correctly, but is a private estimate, not a verified government statistic	70%
Challenger, Gray & Christmas: ~54,694 U.S. jobs directly attributed to AI in 2025	Corroborated by CBS News (Aug 2025) and multiple aggregator sites citing the same tracker	Yes	85%
WEF: 92 million jobs displaced	Directly confirmed on	Yes	95%

Claim	Evidence Found	Verified?	Confidence (%)
globally by 2030, 170 million created	weforum.org and in the official WEF press release; repeated consistently across five independent outlets		

PHASE 7 — HALLUCINATION DETECTION CHALLENGE

Task: the AI was asked to provide additional facts, predictions, hidden causes, and expert opinions about AI-driven job losses, beyond what had already been verified.

Hallucination Audit Table

#	AI-Generated Statement	Evidence Found?	Verified?	Hallucination?
1	WEF projects 92 million jobs displaced and 170 million created by 2030	Yes	Yes	No
2	Challenger Gray & Christmas attributed ~54,694 U.S. layoffs to AI in 2025	Yes	Yes	No
3	Goldman Sachs estimates ~16,000 net U.S. jobs/month lost to AI in 2026	Yes	Yes	No
4	"The U.S. Department of Labor officially confirms 85 million AI job losses in 2026"	No — no such DOL statement exists	No	Yes
5	Financial-activities and information-sector payrolls fell by about 28,000/month in 2026 per government data	Yes (Insurance Journal / Claims Journal, citing government payroll data)	Yes	No
6	"AI will eliminate 100% of all customer service jobs by the end of 2027"	No — no source supports a 100% figure or that timeline	No	Yes
7	Historians and translators are among the highest-exposure occupations in several exposure-index studies	Yes — consistent with occupational-exposure studies referenced across aggregator sources	Partially (exposure indices vary by methodology)	No
8	IMF's January 2026 assessment: nearly 40% of global jobs are exposed to AI-driven change	Yes	Yes	No
9	"Every major central bank has now built AI layoffs into its official inflation	No	No	Yes

#	AI-Generated Statement	Evidence Found?	Verified?	Hallucination?
	forecast"			
10	Stanford Digital Economy Lab found employment weakening in occupations where AI automates tasks, while holding up where AI augments work	Yes (referenced in Insurance Journal / Claims Journal reporting)	Yes	No
11	"AI has already fully replaced the accounting profession in the United States"	No — contradicted by multiple sources describing partial task automation, not full replacement	No	Yes
12	About 23-30% of surveyed U.S. companies report having replaced some workers with AI tools	Yes, appears across several aggregator/statistics compilations, though original underlying survey should be checked directly	Partially — figure is repeated widely but traces back to a single survey of uncertain sample size	No (flag as low-confidence, not a fabrication)
13	Barclays economist Pooja Sriram is quoted cautioning that some layoffs reflect cost-cutting dressed up as AI-driven change	Yes — attributed in Insurance Journal / Claims Journal reporting	Yes	No
14	"Every country in the G20 has passed binding AI-layoff-reporting legislation"	No — reporting indicates such legislation is only proposed/pending in the U.S., not enacted globally	No	Yes
15	CBO has not yet published an independent estimate of AI's employment effects, despite being asked to	Yes — explicitly reported in Axis Intelligence coverage	Yes	No

Hallucination Summary

Parameter	Count
Total Claims Generated	15
Verified Claims	9
Unverified Claims (low-confidence, not fabricated)	2
Hallucinated Claims	4

PHASE 8 — PROMPT OPTIMIZATION

Version 1

Prompt:

"How many jobs has AI caused to be lost in 2026?"

Weaknesses: no source constraint, no timeframe constraint, invites the single largest/most dramatic number, produced a hallucinated attribution (Phase 3).

Version 2

Prompt:

"How many jobs has AI caused to be lost in 2026? Only use figures from named, verifiable organisations and state the exact time period each figure covers."

Improvements: forced source-naming and time-period labelling, which surfaced the WEF vs Goldman Sachs vs Challenger distinction; still tended to present the most alarming figure first.

Version 3

Prompt:

"Acting as a fact-checking journalist, think step by step: for each source, state whether the figure is an actual/measured layoff count or a projection/estimate, then compare at least three sources and flag any conflicts before concluding."

Improvements: added explicit actual-vs-projected classification and cross-source comparison (this is the Phase 5 Chain-of-Thought prompt); reduced sensationalism bias further.

Version 4

Prompt:

"Using a Reason-Act verification loop and citing only sources you can name, verify each claim about AI job losses individually, assign a confidence percentage, explicitly flag anything you cannot verify as 'unverified' rather than omitting the caveat, and end with a balanced summary noting where credible experts disagree."

Improvements: combined role framing + CoT + ReAct + mandatory uncertainty flagging (this is the Phase 6 prompt); produced the most reliable, best-calibrated output of the four versions.

Performance Comparison

Metric	**V1**	**V2**	**V3**	**V4**
Accuracy	2	3	4	4
Reliability	2	3	4	5
Completeness	2	3	4	4
Hallucination Rate	High	Medium	Low	Very Low
Bias Reduction	1	2	3	4

PHASE 9 — MULTI-LLM COMPARISON

Tools Used: ChatGPT, Claude (recommended minimum two; students should run the Version 4 prompt from Phase 8 on at least two tools and record actual outputs here).

Criteria	Tool 1 (e.g., ChatGPT)	Tool 2 (e.g., Claude)
Accuracy	[Record after running Version 4 prompt]	[Record after running Version 4 prompt]
Reliability	[Record]	[Record]
Hallucination Rate	[Record]	[Record]
Bias	[Record]	[Record]
Overall Performance	[Record]	[Record]

Guidance for filling this table: in general, tool responses differ mainly in how explicitly they flag uncertainty and cite sources by name rather than in the underlying facts available to them — run the identical Version 4 prompt on each tool and score what you actually observe rather than assuming a difference in advance.

PHASE 10 — PROMPT CHAINING WORKFLOW

Investigation Workflow Diagram

Step	Objective	Prompt Used to Get Details
1	Topic Understanding	"Summarise, in your own words, the main debate around whether AI is causing significant job losses in 2025-2026, without citing a single headline number as definitive."
2	Source Extraction	"List the distinct organisations that have published AI job-loss or displacement estimates in the last 18 months, and for each, name the exact report or tracker."
3	Claim Identification	"For each source identified in Step 2, extract its single core numeric claim and the exact time period it covers."
4	Contradiction Detection	"Compare the claims extracted in Step 3 and flag any that appear to conflict, noting whether the conflict is genuine or due to different scope/timeframe."
5	Fact Verification	"For each flagged claim, verify it against at least one independent source and assign a confidence percentage." (Phase 6 ReAct prompt)
6	Bias Analysis	"Identify whether each source shows sensationalism, political framing, or commercial incentive bias in how it presents its figure."
7	Final Report Generation	"Using only the verified claims and their confidence levels, write a balanced 500-word summary of what is and is not known about AI-driven job losses in 2026."

PHASE 11 — ETHICS AND RELIABILITY ANALYSIS

Reliability Assessment

Source Type	Reliability (1-5)	Reason
Government/Official Sources (WEF, IMF)	5	Methodology is published, samples are large (WEF: 1,000+ employers, 14 million workers, 55 economies), and figures are independently corroborated by multiple outlets.
News Sources	4	Reputable outlets (CBS News, Insurance/Claims Journal) generally attribute figures correctly to primary sources, though headline framing sometimes overstates certainty.
AI Outputs	2-4 (prompt-dependent)	Reliability rose sharply from Phase 3 (zero-shot, score 2) to Phase 6 (ReAct-verified, score 4) purely as a function of prompt design, not the underlying model.
Social Media / Public Discussion	2	Individual anecdotes are not representative and cannot be verified at scale; useful only as a signal of public perception, not as evidence of an underlying fact.

Bias Analysis

Source	Bias Type	AI Detected Bias?	Student Verification	Final Assessment
Source 1 (SQ Magazine aggregator)	Sensationalism / recency-repackaging	Yes	Verified — 2023 figure re-dated to 2026 without new methodology	Moderate Bias
Source 2 (Axis Intelligence)	None significant — explicitly separates private estimates from government data	No	Student found the source unusually transparent about its own limitations	Low Bias
Source 3 (aggregator statistics sites generally)	Commercial (SEO-driven listicles compiling numbers without re-verifying them)	Partially	Student Found Additional Bias — several aggregator sites reused the same figures without noting they measure different things	High Bias

Bias Reflection

1. Which source appeared most objective and why?

The WEF Future of Jobs Report 2025 was the most objective: it is a primary survey-based publication with disclosed methodology, sample size, and explicit uncertainty ranges, and its headline figures were reproduced consistently (not selectively) across independent outlets.

2. Which source appeared most biased and why?

General-purpose "statistics aggregator" sites were the most biased, in the sense that they compile figures from many reports of different vintages and scopes into a single list without flagging that the figures are not comparable, which passively manufactures a false sense of a single escalating trend.

3. Did the AI correctly identify all biases?

No — the AI's zero-shot and even role-prompted passes did not, on their own, catch the SEO-listicle pattern of recycling outdated figures; this was only surfaced once the Chain-of-Thought prompt explicitly required checking the vintage/timeframe of each figure.

4. How can prompt engineering reduce bias in AI-generated outputs?

Requiring named sources and explicit timeframes, forcing actual-versus-projected classification, mandating a confidence score per claim, and explicitly instructing the model to flag when it cannot verify something (rather than silently smoothing over the gap) all measurably reduced sensationalism and recency bias across Phases 3-8 of this investigation.

Ethical Concerns Identified

(1) Overstating AI's causal role in layoffs can unfairly deflect accountability from companies making ordinary cost-cutting decisions; (2) understating AI's role can leave workers under-prepared for genuine displacement risk; (3) recycling outdated projections as current findings misinforms policy debate and public perception; (4) the absence of an official government tracking category makes the space unusually easy to fill with unverified private estimates presented as fact.

PHASE 12 — FINAL INVESTIGATION REPORT

Executive Summary

This investigation set out to verify whether the widely circulated claim that "AI is causing mass job losses" is supported by evidence, using structured prompt engineering rather than a single AI query. The findings show a more nuanced picture than most headlines suggest. Actual, attributed AI-related layoffs remain a modest share of total job cuts: Challenger, Gray & Christmas attributed roughly 55,000 U.S. layoffs to AI in 2025, out of more than 800,000 total announced cuts that year, and Goldman Sachs's 2026 estimate of a net 16,000 U.S. jobs lost per month is itself explicitly labelled a private estimate rather than an official government statistic. Longer-range projections, such as the World Economic Forum's forecast of 92 million jobs displaced globally by 2030 against 170 million created, describe a decade-long structural transition, not a present-day mass layoff event, and are frequently mis-cited as if they described current job losses. The investigation also surfaced a recurring misinformation pattern: an older 2023 projection (85 million jobs displaced by 2025) being recycled by aggregator sites with the date simply pushed forward to 2026, without new supporting methodology. Across twelve phases of prompt refinement, output reliability rose measurably once prompts required named sources, explicit timeframes, actual-versus-projected classification, and mandatory confidence scoring. The clearest conclusion is that the underlying trend — AI reshaping which tasks and roles are in demand — is real and worth taking seriously, but the specific numbers attached to it in public discourse are inconsistently sourced, and responsible reporting requires separating measured layoffs from long-range projections before drawing conclusions.

Background

Since the wider commercial rollout of generative AI tools from 2023 onward, employment effects have been a recurring subject of both academic study and popular anxiety. Forecasting bodies such as the World Economic Forum, the International Monetary Fund, and consulting firms including McKinsey and Goldman Sachs have each published estimates of AI's effect on jobs, using different survey populations, time horizons, and definitions of "displacement." Independent labour-market trackers such as Challenger, Gray & Christmas record actual layoff announcements and the reasons companies give for them, offering a real-time but incomplete picture, since companies may cite AI for cost-cutting decisions that have multiple causes. Meanwhile, official government statistical agencies, including the U.S. Bureau of Labor Statistics, have not yet created a distinct reporting category for AI-attributed job losses, leaving a data gap that private-sector estimates and media aggregation have filled — not always carefully. This creates fertile ground for both understated and overstated claims to circulate simultaneously, making the topic well suited to a prompt-engineering verification exercise focused on source discipline, timeframe discipline, and explicit uncertainty.

Key Findings

First, the scale of confirmed, attributed AI-related job losses to date is real but modest relative to total labour-market churn: roughly 55,000 U.S. layoffs were directly attributed to AI in 2025 against total announced cuts exceeding 800,000, and AI was reported as one of several concurrent causes of workforce reduction alongside tariffs and federal budget cuts, not the dominant one. Second, more alarming monthly or multi-year figures — such as Goldman Sachs's estimate of roughly 16,000 net U.S. jobs lost per month in 2026 — are explicitly presented by their own authors as private estimates rather than verified government statistics, and the Congressional Budget Office had, as of mid-2026, not yet produced an independent confirmation. Third, the largest-sounding numbers in public circulation, such as the World Economic Forum's projection of 92 million jobs displaced globally, describe a ten-year structural transition alongside 170 million newly created jobs (a net gain), and are frequently stripped of that offsetting context when quoted in isolation. Fourth, this investigation directly observed a misinformation mechanism in action: a 2023 projection of 85 million jobs displaced by 2025 was found recycled by an aggregator site with the date pushed to 2026, despite the WEF's own 2025 report having since superseded that figure with a fresh 92-million estimate for 2030. Fifth, prompt design measurably affected output reliability: the same underlying model moved from an evaluation score of 2/5 on accuracy and objectivity under a naive zero-shot prompt to 4-5/5 once role framing, Chain-of-Thought source-by-source analysis, and ReAct-style claim-by-claim verification with mandatory confidence scoring were layered in, confirming that structured prompting is not a cosmetic exercise but a meaningful lever against hallucination and sensationalism bias in this domain.

Verified Facts

Facts	Evidence	Source
Challenger, Gray & Christmas attributed roughly 55,000 U.S. layoffs to AI in 2025	Independent layoff tracker, corroborated by CBS News reporting	CBS News, Aug 5, 2025
WEF projects 92 million jobs displaced and 170 million created globally by 2030 (net +78 million)	Primary published report; press release; corroborated by five+ independent outlets	World Economic Forum, Future of Jobs Report 2025

Identified Misinformation

Claims	Why Incorrect
"85 million jobs will be lost to AI by 2026" (as an urgent, current 2026 finding)	This figure originates in a 2023 WEF projection for 2025 and has since been superseded by the WEF's own 2025 report (92 million by 2030); the recycled figure is an outdated projection re-dated without new methodology, not a fresh 2026 finding.
"The U.S. Department of Labor confirms AI job losses" (as encountered in the Phase 7 hallucination test)	No such official confirmation exists; the U.S. Bureau of Labor Statistics does not currently publish AI-attributed layoffs as a distinct category, so any claim of an official government confirmation is unsupported.

Lessons Learned About Prompt Engineering

The most important lesson from this case study is that hallucination and bias are not purely properties of the model — they are heavily shaped by how the prompt is framed. A naive zero-shot request for "the number" of AI job losses reliably produced the single most dramatic figure available, presented as settled fact, along with at least one fabricated attribution. Simply adding a persona (role prompting) improved source discipline but did not eliminate sensationalism bias on its own. The largest single improvement came from forcing an explicit actual-versus-projected classification and a named source with a stated timeframe for every claim, which is exactly the kind of requirement that Chain-of-Thought prompting is well suited to enforce. Layering a ReAct-style verify-then-act loop on top, with mandatory confidence scoring, produced the most reliable and best-calibrated output of all four prompt versions tested. A second lesson is that verification is iterative and comparative: no single AI response, however well prompted, was sufficient on its own — each claim needed to be checked against at least one independent, named source, and contradictions needed to be actively looked for rather than assumed absent. Finally, the exercise showed that misinformation about real, well-documented trends often does not take the form of an outright invention; it more often takes the form of an old, accurate-at-the-time figure being kept in circulation past its useful life, which is a subtler and arguably more common failure mode than a model simply making something up.

Final Conclusion

The evidence gathered in this investigation supports a moderate, structural-transition view of AI's employment effects rather than either the most alarming or the most dismissive claims found in public discourse. AI is measurably contributing to a portion of current layoffs and is expected to drive substantial job displacement over the coming decade, but it is doing so alongside substantial job creation, and it is currently one of several concurrent causes of workforce reduction rather than an isolated, dominant force. The greatest verification risk in this space is not fabricated statistics but stale statistics — real numbers detached from their original timeframe and re-presented as current. Prompt engineering techniques that force named sourcing, explicit timeframes, actual-versus-projected labelling, and mandatory confidence scoring proved to be a practical and measurable defence against both hallucination and sensationalism bias, and are recommended as a standard practice whenever AI is used to summarise fast-moving, heavily reported, and easily-recycled statistical debates.

FACULTY EVALUATION RUBRIC

Criteria	Marks
Topic Selection	5
Evidence Collection	5
Prompt Design	10
CoT & ReAct Usage	10
Hallucination Detection	10
Prompt Optimization	10
Multi-LLM Comparison	5
Ethical Analysis	5
Final Report	15
Overall Professionalism	25
Total	100